SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 1 **Total Marks:10**

Annexure A

Descriptive Written Test

Code of Conduct:

*I **B Reddy Balu (192312499)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Balu
Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Question 1: Intelligent Agents in a Smart Home Automation System

1. Introduction

An intelligent agent is anything that perceives its environment through sensors and acts upon that environment through actuators in order to achieve its objectives. The rational agent is the one that selects the action expected to maximize its performance measure, given the percept sequence it has seen and its built-in knowledge. A smart home automation system is a good example of an intelligent agent, because it continuously senses the home and changes the lights, heating/cooling and security devices on its own.

2. Problem Formulation (PEAS Description)

Before choosing an agent type, the task environment of the smart home is specified using the PEAS framework.

Component	**Smart Home Description**
Performance measure	Occupant comfort, low energy consumption/cost, fast and reliable detection of security threats, few false alarms, respect for user preferences.
Environment	Rooms of a house, occupants (family members, guests, intruders), weather, time of day, electricity tariff, appliances.
Actuators	Smart bulbs/dimmers, thermostat, AC/heater, fans, smart door locks, alarm siren, camera recording, mobile notifications.
Sensors	PIR motion sensors, temperature and humidity sensors, light (LUX) sensors, door/window contact sensors, cameras, smart clock, smart meter, user app input.

Properties of the environment: it is partially observable (sensors cannot see everything, e.g. a person in an unmonitored room), stochastic (occupant behaviour and weather are unpredictable), sequential (current actions affect future comfort and energy use), dynamic (it changes while the agent is deciding), continuous (temperature, light level, time) and multi-agent (several occupants with different preferences). These properties explain why a single, very simple agent design is not enough.

3. Types of Intelligent Agents and Their Role in the Smart Home

3.1 Simple Reflex Agent

Concept: A simple reflex agent selects actions using only the current percept, ignoring the percept history. It works through condition–action rules (if condition then action). It has no memory and no model of the world, so it works properly only when the environment is fully observable and the correct action can be decided from the current input alone.

Working in the smart home: Rules are triggered directly by sensor readings. Examples:

- IF motion detected AND room is dark THEN turn the light ON.
- IF temperature > 30°C THEN switch the AC/fan ON.
- IF door sensor opened AND system is in "Armed" mode THEN sound the alarm.
- IF smoke or gas sensor value exceeds the limit THEN trigger the alarm and unlock exits.

Advantages: Very fast, cheap, simple to program and highly reliable for safety-critical emergency reflexes such as fire or gas alarms.

Limitations: It cannot remember anything. It may switch lights off when someone sits still (no motion for a moment), keep switching the AC on and off around the threshold, cannot distinguish a family member from an intruder, and cannot learn user habits or save energy intelligently.

3.2 Model-Based Reflex Agent

Concept: A model-based reflex agent maintains an internal state that tracks the part of the world it cannot see now. It uses a model of how the world evolves and how its own actions change the world. The action is chosen by condition–action rules applied to this updated internal state, so it can work in partially observable environments.

Working in the smart home: The agent stores which rooms are occupied, when motion was last detected in each room, whether the front door is locked, the last thermostat setting and whether windows are open. Examples:

- If motion in the living room stopped only 2 minutes ago and the person was seated, the agent infers that the room is probably still occupied and keeps the light on.
- If the agent remembers that the front door was opened and no known occupant is inside, it infers a possible intrusion.
- It knows that heating a room takes time, so it does not repeatedly toggle the AC.
- It knows that a closed window plus a running AC will cool the room faster, and updates its belief about room temperature even between sensor readings.

Advantages: Handles partial observability, avoids unstable on/off switching, gives smoother and more realistic behaviour than reflex agents.

Limitations: It still only reacts to the current state through fixed rules. It has no explicit objective, so it cannot decide between competing options (for example comfort versus saving energy) or plan ahead.

3.3 Goal-Based Agent

Concept: A goal-based agent has, in addition to a world model, explicit goal information that describes desirable situations. It chooses actions by considering the future: it uses search and planning to find an action sequence that reaches the goal. It is more flexible, because changing the goal changes the behaviour without rewriting all rules.

Working in the smart home: Goals such as "room temperature at 24°C when the user reaches home at 6 PM" or "house fully secured before bedtime at 11 PM" are given.

- To reach 24°C by 6 PM, the agent plans to start the AC at 5:40 PM (pre-cooling), based on the current temperature and the model of the cooling rate.
- For the goal "house secured", it plans the sequence: check all windows, lock doors, switch off unnecessary lights, arm the alarm, and turn on cameras.
- For "Vacation mode", it plans to simulate occupancy by turning lights on and off at random evening hours.

Advantages: Proactive and planning-oriented; easy to modify by changing goals; can handle multi-step tasks.

Limitations: Goals give only a binary distinction between "achieved" and "not achieved". When several action sequences achieve the goal, a goal-based agent cannot tell which one is better (cheaper, quicker, more comfortable), and it cannot trade off conflicting goals.

3.4 Utility-Based Agent

Concept: A utility-based agent uses a utility function that maps each state (or outcome) to a real number representing the degree of happiness or desirability. It selects the action with maximum expected utility, taking into account uncertainty and trade-offs between conflicting objectives. It is therefore the most general rational agent.

Working in the smart home: A utility function can combine several factors, for example:

U(state) = w1 × Comfort + w2 × Energy Savings + w3 × Security − w4 × Cost

- Cooling: the agent chooses between AC at 22°C (high comfort, high energy cost), AC at 26°C with a fan (slightly lower comfort, much lower cost), or no cooling during peak tariff. It selects the option with the highest overall utility, perhaps pre-cooling during cheap off-peak hours.
- Lighting: it balances natural daylight, user brightness preference and electricity use by dimming bulbs rather than switching them fully on.
- Security: it weighs the cost of a false alarm (annoyance, neighbour disturbance) against the cost of a missed intrusion, and decides whether to only notify the owner or sound the siren.
- Multiple occupants: when two people prefer different temperatures, it chooses the setting that maximizes total (or fair) utility.

Advantages: Handles trade-offs and uncertainty, gives the best-quality decisions, and can be improved with learning of the weights from user feedback.

Limitations: Designing a correct utility function is difficult, and computing expected utility is computationally more expensive than simple rules.

4. Comparative Analysis

Aspect	**Simple Reflex**	**Model-Based**	**Goal-Based**	**Utility-Based**
Uses	Current percept only	Percept + internal state	State + explicit goals	State + utility function
Memory/model	None	Yes (world model)	Yes + goals	Yes + preferences and uncertainty
Decision basis	Condition–action rules	Rules on internal state	Search / planning	Maximum expected utility
Smart home example	Motion → light ON	Keeps light ON for seated person	Pre-cool house before arrival	Best comfort–cost trade-off
Comfort	Low	Medium	Medium–High	High
Energy efficiency	Low (wasteful)	Medium	Medium–High	High (optimized)
Security	Good for instant alarms; many false alarms	Better; tracks door/occupancy state	Good; plans lock-down routines	Best; balances false alarm vs missed threat

Aspect	Simple Reflex	Model-Based	Goal-Based	Utility-Based
Speed	Fastest	Fast	Slower (planning)	Slowest (evaluation)
Can learn?	No	Limited	Limited	Yes, weights can be learned

5. Most Effective Approach: Hybrid Layered Agent with Learning

No single agent type satisfies all the requirements of the smart home. Simple reflex agents are fast but ignorant, while utility-based agents are intelligent but slower. The most effective solution is therefore a hybrid (layered) architecture that combines all four types and adds a learning component, so that each layer is used where it is strongest.

Layer	Agent type	Responsibility in the smart home
Layer 1: Reactive (highest priority)	Simple reflex	Instant safety reflexes: smoke/gas alarm, glass break or forced-entry siren, emergency light when motion occurs in the dark. These bypass slower layers for guaranteed fast response.
Layer 2: State tracking	Model-based	Maintains the house state: room occupancy, door/window status, appliance status, temperature trend. Filters noisy sensor data and prevents rapid on/off switching.
Layer 3: Planning	Goal-based	Achieves scheduled goals such as pre-cooling, bedtime security routine, vacation mode and "welcome home" scenes.
Layer 4: Optimization	Utility-based	Resolves conflicts and trade-offs between comfort, energy cost and security when choosing among several valid actions.
Learning element	Learning agent component	Learns user habits (wake-up time, preferred temperature, usual arrival time) from feedback and history, and updates the model, goals and utility weights over time.

6. Justification

Comfort: The model-based layer knows who is in which room, and the goal-based layer pre-adjusts temperature and lighting before the user needs them. The utility layer personalizes the settings, and the learning element adapts them to changing preferences (for example, a warmer setting in winter evenings). This makes the home feel responsive instead of merely reactive.

Energy efficiency: The model-based layer switches off devices in unoccupied rooms and avoids wasteful on/off cycling. The utility-based layer chooses actions that cost less (using daylight, dimming lights, running pre-cooling in off-peak tariff hours, setting AC a few degrees higher with a fan) while keeping comfort at an acceptable level. Learning the occupancy schedule further reduces energy waste.

Security: The reactive layer guarantees an immediate alarm for critical events, with no planning delay. The model-based

layer recognizes abnormal situations (door opened while everyone is away) and the utility layer reduces false alarms by weighing evidence from multiple sensors (motion + door + camera + time of day) before escalating. The goal-based layer automates the nightly "secure the house" routine.

Real-time response and scalability: Because time-critical decisions are handled by the fast lower layers while the planning and optimization layers work on slower, non-urgent decisions, the system satisfies real-time constraints. New devices or rules can be added to a particular layer without redesigning the whole system.

Adaptation: The learning component allows the system to improve with experience, which a fixed rule-based agent cannot do. This meets the requirement that the system should learn user preferences over time.

7. Example Scenario

At 7:00 PM on a hot day, the user is returning home. The goal-based layer has already started pre-cooling at 6:40 PM, using the learned arrival time. The utility layer chose 25°C with the ceiling fan, since this gives almost the same comfort as 22°C but at much lower energy cost. As the user enters, the motion sensor (reflex layer) turns the entrance light on immediately. The model-based layer records that the user is home, disarms the security system for that user and marks the living room as occupied. At 11 PM, the "bedtime" goal triggers locking of doors, switching off unused lights and arming of the alarm. If a window sensor then reports forced opening, the reflex layer sounds the siren at once, and a notification is sent to the owner.

8. Conclusion

Simple reflex agents give speed, model-based agents give awareness of the hidden state, goal-based agents give planning ability, and utility-based agents give optimal trade-offs between conflicting objectives. Because a smart home must react in real time, operate under partial observability, balance comfort, energy and security, and learn user behaviour, a hybrid layered agent with a learning component is the most effective approach. This design gives fast and safe responses, intelligent energy-saving behaviour and personalized comfort, which a single agent type cannot provide alone.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand